



Tshwane University of Technology

We empower people

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NAME OF SYSTEM

Real Estate Management System

OBJECTIVES/VISION

This project is aimed at developing a real estate management system that is a collection of registers and reports for the effective management of properties. Besides this, administrators and agents can view property listings, tenant records, and transaction reports for their purpose. The management of properties includes property registration, tenant admission, property sales, rental agreements, maintenance requests, and suitable reports regarding these aspects.

USERS OF THE SYSTEM

- Property Manager
- Property Owner
- Maintenance Staff
- Tenant
- Administrator
- Real Estate Agent

FUNCTION REQUIREMENTS

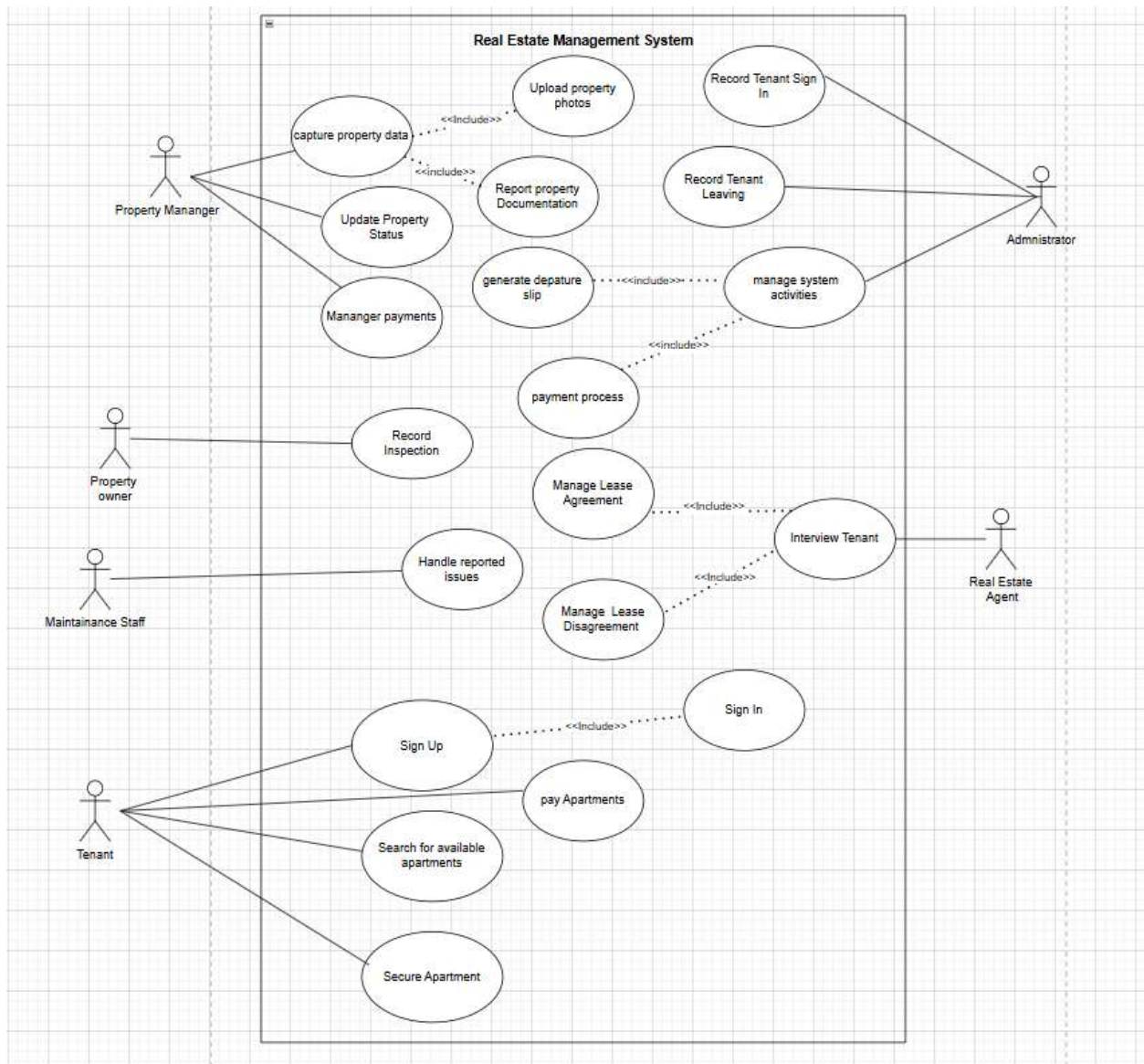
- User registration and authentication
- Property data capture and updates
- Photo and document management
- Tenant sign-in and leaving processes
- Lease management and dispute handling
- Payment processing
- Inspections and maintenance tracking
- Apartment search and reservation
- Tenant screening
- Administrative monitoring

Event	Trigger	Source	Use Case	Response	Destination
Buyer applies for property	Buyer submits property application	Buyer	Property application	Application recorded	Database/Admin

Administrator reviews application	Administrator opens pending applications	Administrator	Review application	Application approved/rejected	Buyer/Admin
Tenant submits rental application	Tenant submits rental form	Tenant	Rental application	Application recorded	Database/Admin
Administrator reviews rental app	Administrator checks rental applications	Administrator	Review application	Rental approved/rejected	Tenant/Admin
Property Agent captures property	Agent enters property details	Property Agent	Capture property data	Property record created	Database
Agent uploads property photo	Agent attaches photo to property record	Property Agent	Upload property photo (extends)	Photo linked to property record	Database
Tenant admitted to property	Tenant assigned to property	Data Clerk	Record tenant submission	Tenant record updated	Database

Event	Trigger	Source	Use Case	Response	Destination
Property sale recorded	Buyer completes purchase	Property Agent	Record property sale	Sale transaction logged	Database/Admin
Tenant requests maintenance	Tenant submits maintenance request	Tenant	Record maintenance request	Request logged	Database/Admin
Administrator generates property report	Administrator requests property report	Administrator	Print property report	Property report generated	Administrator
Administrator generates tenant report	Administrator requests tenant report	Administrator	Print tenant report	Tenant report generated	Administrator
Administrator generates sales report	Administrator requests sales report	Administrator	Print sales report	Sales report generated	Administrator
Administrator generates maintenance report	Administrator requests maintenance report	Administrator	Print maintenance report	Maintenance report generated	Administrator

USE CASE DIAGRAMS



USE CASE DESCRIPTION

Use case name:	Capture Property Data	
Triggering event:	A Property Manager decides to add a new property listing into the system.	
Brief description:	A real estate agent records complete property information (owner details, property specifications, pricing, and images) into the system so that the property can be listed for sale or rent.	
Actors:	Property Manager.	
Related use cases:	Validate Owner Details; Upload Property Images; Verify Property Information.	
Pre-conditions:	• Property owner has submitted the property for listing. • The agent is authenticated and logged into the system. • The property does not already exist in the system. • Required property documentation is available.	
Post-conditions:	• Property record is successfully stored in the system database. • Property becomes available for search and listing. • Confirmation is displayed to the agent.	
Flow of activities:	ACTOR	System
	1. Selects “Add New Property” option. 2. Enters property details (address, type, size, price, description). 3. Enters property owner information. 4. Uploads property images and supporting documents. 5. Submits the property listing.	1.1 Displays property data entry form. 2.1 Validates mandatory fields and data format. 3.1 Verifies owner details (via <i>Validate Owner Details</i> use case). 4.1 Stores images and documents in the system. 5.1 Saves property data to database. 5.2 Generates property ID. 5.3 Displays confirmation message.

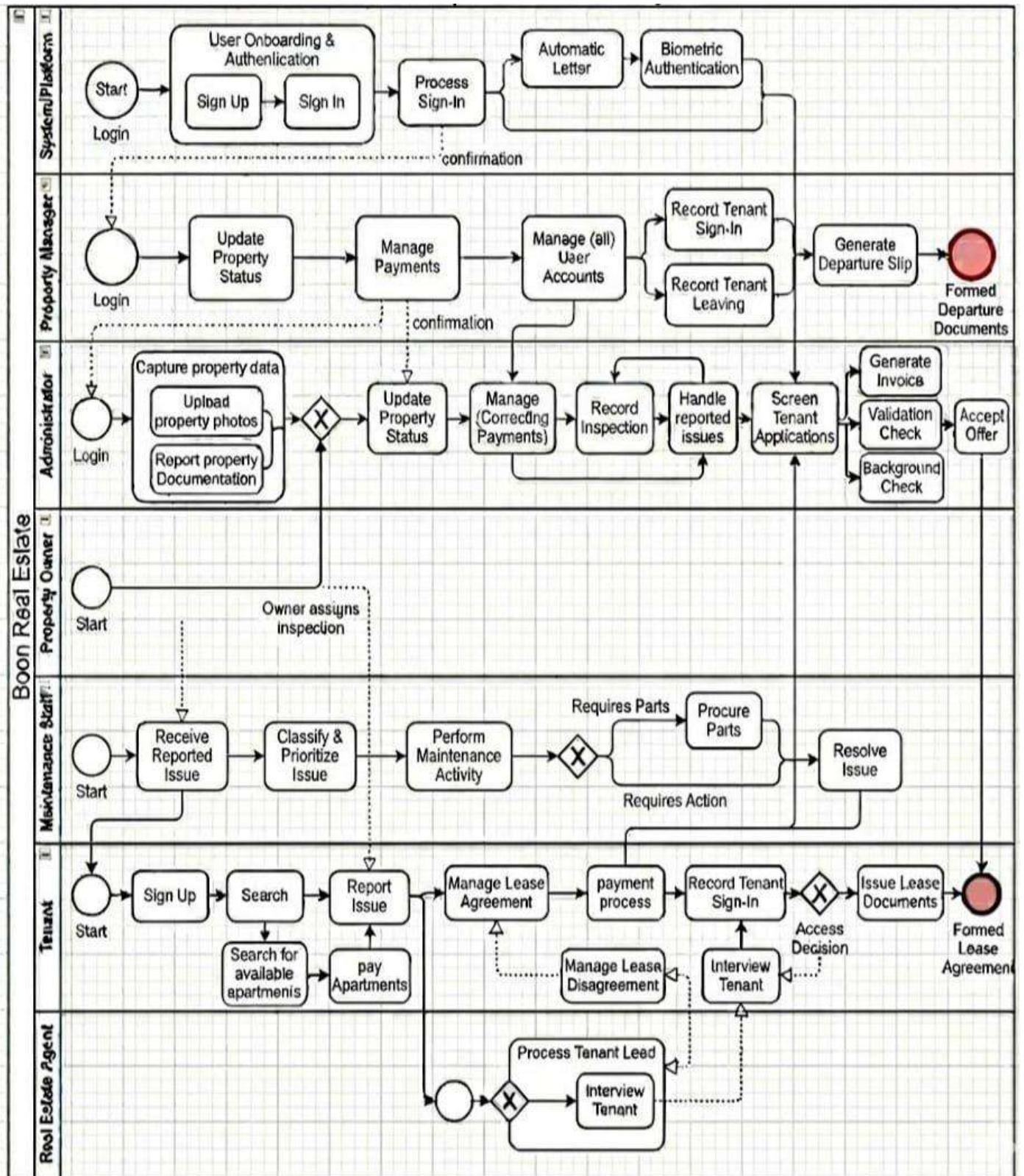
Use case name:	Handle Reported Issues
Triggering event:	Maintenance Staff logs into the system to view and address pending maintenance requests submitted by Tenants.
Brief description:	The Maintenance Staff reviews reported maintenance problems, updates the status of repairs, and records the resolution details so that the Property Manager and Tenant are kept informed.
Actors:	Maintenance Staff.
Related use cases:	Update Property Status; Record Inspection; Manage Lease Disagreement.

Pre-conditions:	• Maintenance Staff is authenticated and logged into the system. • At least one maintenance issue has been reported by a Tenant. • The property associated with the issue exists in the system.	
Post-conditions:	• The status of the issue is updated (e.g., In Progress, Resolved). • Maintenance logs are successfully stored in the system database. • Notification of progress is sent to the relevant Tenant/Manager.	
Flow of activities:	ACTOR	System
	1. Selects "View Reported Issues" from the dashboard. 2. Selects a specific issue to view detailed descriptions and photos. 3. Updates the status of the issue to "In Progress" or "Scheduled." 4. Enters maintenance notes, costs (if applicable), and resolution steps taken. 5. Marks the issue as "Resolved" and submits the final report.	1.1 Retrieves and displays a list of all active maintenance tickets. 2.1 Displays the full issue report, including property location and tenant contact info. 3.1 Validates the status change and updates the record. 4.1 Captures and stores maintenance activity data. 5.1 Saves updated status to the database. 5.2 Generates a maintenance completion log. 5.3 Displays confirmation message and notifies the Tenant.

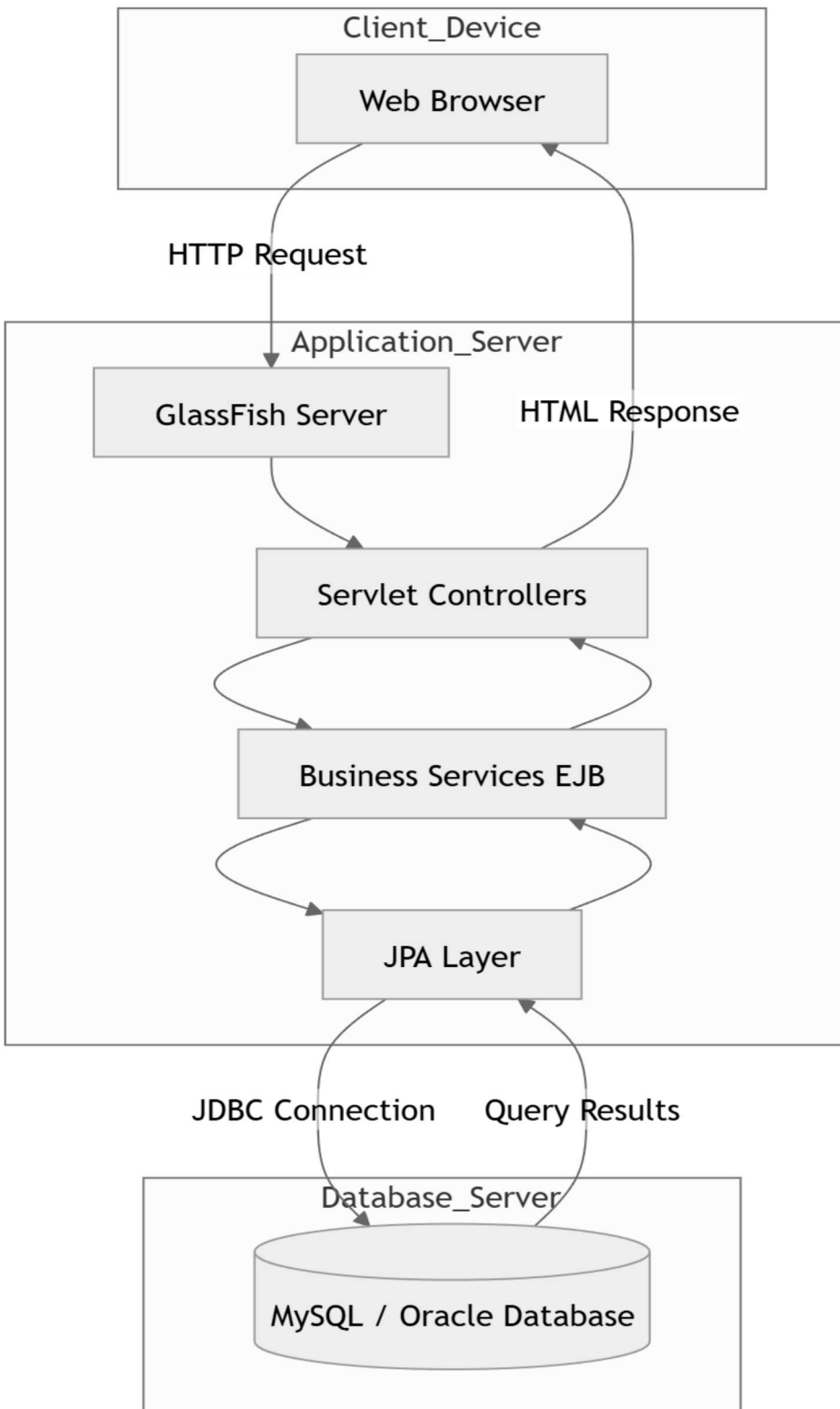
Use case name:	Manage System Activities	
Triggering event:	An Administrator needs to monitor system logs, review financial transactions, or generate operational reports.	
Brief description:	The Administrator manages the overall operations of the Real Estate Management System, which includes overseeing payment processes and generating departure slips.	
Actors:	Administrator	
Related use cases:	Include: Generate departure slip; payment process.	
Pre-conditions:	• The Administrator is authenticated and logged into the system. • The system is active and capturing data from other actors (Tenants, Managers).	
Post-conditions:	• System activity logs are updated. • Operational documents or payment records are successfully processed.	
Flow of activities:	ACTOR	System

	1. Selects "Manage System Activity" from the administrative dashboard. 2. Initiates a review of the current payment process 3. Selects the option to generate departure slip for a leaving tenant. 4. Reviews system logs or activity reports for errors. 5. Confirms and saves the administrative actions.	Displays the system management interface with options for payments and slips. 2.1 Retrieves and displays pending or completed transaction records. 3.1 Pulls tenant data and Generates the formal departure documentation. 4.1 Filters and Displays requested activity logs. 5.1 Saves activity history to the database and Displays confirmation.
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BUSINESS PROCESS DIAGRAM



BLOCK DIAGRAM



PLANNING [WORK BREAKDOWN STRUCTURE]

First space			
Boon RealEstate Info Copilot			
Table Board Gantt chart Files Analytics Calendar + All tasks			
	Name ↑	Status	Due date
1	▼ Boon RealEstate	In Progress	10/02/2026
2	▼ 1. Initiation	Completed	13/02/2026
3	Define scope & objectives	Completed	10/02/2026
4	Gather requirements	Completed	12/02/2026
5	▼ 2. Design	Completed	20/02/2026
6	Application architecture (D...	Completed	18/02/2026
7	Database schema design	Completed	19/02/2026
8	UML diagrams (use cases, ...	Completed	24/02/2026
9	▼ 3. Development	In Progress	11/03/2026
10	DAO classes (Property, Ten...	Completed	04/03/2026
11	GUI/Servlet mappings	In progress	06/03/2026
12	Implement DatabaseConn...	New	09/03/2026
13	Reports module	New	11/03/2026
14	▼ 4. Testing	New	18/03/2026
15	GUI testing & error handli...	New	12/03/2026
16	Integration testing (Derby ...	New	13/03/2026
17	Unit testing DAO methods	New	16/03/2026
18	User acceptance testing	New	18/03/2026
19	▼ 5. Deployment	New	25/03/2026
20	Configure environment va...	New	20/03/2026
21	Deploy to GlassFish/NetBe...	New	23/03/2026
22	Package WAR file	New	25/03/2026
23	▼ 6. Project Management	New	02/04/2026
24	Gantt chart updates	New	27/03/2026
25	Task assignment & tracking	New	02/04/2026
+	Item		

